

CS714: Secure Computation

Adithya Vadapalli

IIT Kanpur

February 17, 2026

A Three-Way Coin Toss

Goal

- 1 There are three parties P_0 , P_1 , and P_2 .
- 2 They would like to securely do a three-way toss among them.

Additive Secret Shares

Shares over a Ring $\mathbb{Z}_{2^{64}}$

If P_0 holds x_0 , P_1 holds x_1 , ..., P_n holds x_n such that $(x_0 + x_1 + \dots + x_n) \bmod \mathbb{Z}_{2^{64}} = x$. Then we say each party holds additive shares of x .

XOR Secret Shares

Bitwise XOR Shares over $\mathbb{F}_2$

Let $x \in \{0, 1\}^\ell$ be a bitstring. If parties $P_0, P_1, \dots, P_n$ hold shares $x_0, x_1, \dots, x_n \in \{0, 1\}^\ell$ such that

$$x = x_0 \oplus x_1 \oplus \dots \oplus x_n,$$

then we say they hold *XOR shares* of x .

Why This Is Sharing over a Field

XOR is addition in the finite field $\mathbb{F}_2$:

$$0 \oplus 0 = 0, \quad 0 \oplus 1 = 1, \quad 1 \oplus 1 = 0.$$

Thus, each bit of x is additively shared over $\mathbb{F}_2$. An ℓ -bit XOR sharing is equivalent to ℓ independent additive sharings in $\mathbb{F}_2$.

Example: Secret Sharing over $\mathbb{F}_8$

The Field $\mathbb{F}_8$

$\mathbb{F}_8$ is a finite field with $8 = 2^3$ elements. It can be represented as:

$$\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$$

Each field element corresponds to a 3-bit vector.

Additive Secret Sharing over $\mathbb{F}_8$

Let $x \in \mathbb{F}_8$. If parties $P_0, P_1, \dots, P_n$ hold shares $x_0, x_1, \dots, x_n \in \mathbb{F}_8$ such that $x = x_0 + x_1 + \dots + x_n$ (over $\mathbb{F}_8$), then they hold additive shares of x over $\mathbb{F}_8$.

Connection to XOR Sharing

Since $\mathbb{F}_8$ is a vector space over $\mathbb{F}_2$, addition in $\mathbb{F}_8$ corresponds to bitwise XOR on 3-bit representations. Thus, sharing in $\mathbb{F}_8$ is equivalent to three correlated XOR sharings over $\mathbb{F}_2$.

Finite Fields of Size 2^k

Finite Fields

A finite field $\mathbb{F}_q$ is a set with:

- Addition and multiplication
- Additive and multiplicative inverses
- No zero divisors

Such a field exists *iff* $q = p^k$ for a prime p .

The Field $\mathbb{F}_8$

$$\mathbb{F}_8 = \mathbb{F}_{2^3}$$

It has 8 elements and characteristic 2. Each element can be represented as a polynomial:

$$a_0 + a_1\alpha + a_2\alpha^2 \quad (a_i \in \mathbb{F}_2)$$

with arithmetic done modulo an irreducible polynomial, e.g.

$$\alpha^3 + \alpha + 1 = 0.$$

From $\mathbb{F}_8$ to XOR Shares

Vector Space View

$\mathbb{F}_8$ is a 3-dimensional vector space over $\mathbb{F}_2$. Each element corresponds to a 3-bit vector:

$$a_0 + a_1\alpha + a_2\alpha^2 \longleftrightarrow (a_0, a_1, a_2)$$

Addition in $\mathbb{F}_8$

Addition is coefficient-wise over $\mathbb{F}_2$: $(a_0, a_1, a_2) + (b_0, b_1, b_2) = (a_0 \oplus b_0, a_1 \oplus b_1, a_2 \oplus b_2)$

Thus, **addition in $\mathbb{F}_8$ is bitwise XOR**.

Implication for Secret Sharing

Additive sharing over $\mathbb{F}_8$ is equivalent to

- three coupled XOR sharings over $\mathbb{F}_2$
- but with meaningful *field multiplication*

What Does $\mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$ Mean?

Quotient Construction

$\mathbb{F}_2[\alpha]/(f(\alpha))$ means:

- work with polynomials over $\mathbb{F}_2$
- identify polynomials that differ by a multiple of $f(\alpha)$
- equivalently: reduce modulo $f(\alpha)$

For $\mathbb{F}_8$

Using $f(\alpha) = \alpha^3 + \alpha + 1$: $\alpha^3 = \alpha + 1$ Every element has a unique remainder: $a_0 + a_1\alpha + a_2\alpha^2$

Analogy

$\mathbb{Z}/7\mathbb{Z}$: integers mod 7 $\mathbb{F}_2[\alpha]/(f)$: polynomials mod f

What Does $\alpha^3 = \alpha + 1$ Mean?

Defining Relation

In the quotient

$$\mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1),$$

the polynomial $\alpha^3 + \alpha + 1$ is defined to be zero.

Rewrite Rule

Thus, inside the field: $\alpha^3 = \alpha + 1$ This is not an identity, but a reduction rule.

Analogy

In $\mathbb{Z}/7\mathbb{Z}$: $7 = 0 \Rightarrow 8 = 1$ Same idea, different objects.

Why Finite Fields Have Size p^k

Characteristic

In any finite field F , repeated addition of 1 must cycle. The smallest p such that

$$p \cdot 1 = 0$$

is the characteristic of F .

Why p Must Be Prime

If $p = ab$, then:

$$(a \cdot 1)(b \cdot 1) = 0$$

which would create zero divisors. Hence p is prime.

Vector Space View

Every finite field F contains $\mathbb{F}_p$ and is a k -dimensional vector space over it. Thus:

A Multiplication Protocol

Why $\mathbb{Z}_{2^k}$ Is Not a Field

Recall: What a Field Requires

A field has:

- no zero divisors
- multiplicative inverses for all nonzero elements

Zero Divisors in $\mathbb{Z}_{2^k}$

In $\mathbb{Z}_{2^k}$:

$$2 \cdot 2^{k-1} = 2^k \equiv 0 \pmod{2^k}$$

but neither factor is zero.

Concrete Example

In $\mathbb{Z}_8$:

$$2 \cdot 4 = 8 \equiv 0$$

Hence, 2 and 4 are zero divisors.

What Does “No Zero Divisors” Mean?

Definition

A ring has *no zero divisors* if:

$$a \cdot b = 0 \Rightarrow a = 0 \text{ or } b = 0$$

Bad: $\mathbb{Z}_8$

$$2 \cdot 4 = 8 \equiv 0$$

with $2 \neq 0$, $4 \neq 0$. Thus, 2 and 4 are zero divisors.

Why Fields Care

Zero divisors prevent multiplicative inverses. Hence, fields cannot have zero divisors.

Why Zero Divisors Break Inverses

Assume an Inverse Exists

Let $a \neq 0$ and suppose a^{-1} exists.

Zero Divisor Assumption

If $a \cdot b = 0$ for some $b \neq 0$, then:

$$a^{-1}(a \cdot b) = 0$$

$$(a^{-1}a)b = b = 0$$

Contradiction.

Conclusion

An element with a multiplicative inverse cannot be a zero divisor.